CHAPTER 4

The Scientific Revolution

On October 11, 1572, the Danish astronomer Tycho Brahe (1546–1601) observed a bright new object in the evening sky. This was "a miracle indeed," since this object did not move against the background of fixed stars and must itself have been a star. Yet Aristotle had taught that everything in the celestial region, the sphere of fixed stars, was perfect and unchanging. Brahe's observation of what we now believe to have been a supernova (a new star) was one of the many developments that led to the eventual overthrow of the Aristotelian geocentric (Earth-centered) astronomical system and the medieval worldview based upon it.

By the 14th century, the social, political, and intellectual order of the medieval world had begun to break down. Increased urbanization and the return to a money economy eroded the structure of the feudal system, and the rise of nation-states undermined the political authority of the papacy. Intermittent wars between the emerging nation-states led to a severe economic depression. This was followed by the plague of 1348–1350, later known as the "Black Death," which decimated the European population and bred doubt and resentment against the medieval Church, the dominant authority. Although the Church embraced Aristotle's philosophy, the threat posed by its naturalism and rationalism generated dissent and division, leading initially to attempts to divorce the separate realms of faith and reason and then to the autonomous emergence of naturalistic empirical science.

Various developments contributed to the transformation of the intellectual landscape. Marco Polo's (1254–1324) exploration of China, Christopher Columbus's (1451–1506) discovery of America in 1492, and Magellan's (1480–1521) circumnavigation of the globe expanded the horizons of the known world. Perhaps the most significant development was the invention of printing and the consequent transformation of communication. In the city of Mainz in southern Germany, Johann Gutenberg (c. 1397–1468) created movable type and published an edition of the Bible in 1450. The consequent explosion in printed works expanded intellectual horizons by broadening access to the Bible and classical works. By 1500, about 8 million volumes had been printed (Pyenson & Sheets-Pyenson, 1999); by 1600, about 20 million, with over a dozen presses established in European cities (Foote, 1991). The critical interpretation of these works by humanist scholars encouraged a more secular—and more skeptical—approach to the classical tradition and scriptural authority, and the reliable reproduction of works in physics, astronomy, and medicine transformed science into a public enterprise. In earlier centuries the works of Aristotle, Ptolemy, and Galen had been transcribed by hand by monastic clerics, with errors compounded over generations, and read only by the educated elite. From the mid-15th century onwards multiple copies of scientific works were critically scrutinized by the scientific community and educated members of the lay public.

Critical questioning of the classical tradition and scriptural authority was paralleled by the critical and empirical evaluation of the theories of Aristotle, Ptolemy, and Galen during the period of the 16th and 17th centuries in Europe known as the scientific revolution. As their theories were displaced by those of Nicholaus Copernicus (1473–1543), Johannes Kepler (1571–1630), Galileo, Newton, and Vesalius, empirical evaluation displaced the authority of tradition as the mark of modern science. Quantified efficient causal explanation of matter in motion displaced final causal explanation in the new physics, and eventually these mechanistic forms of explanation were extended to the realm of biology and psychology by theorists such as Gomez Pereira (1500–c. 1558), William Harvey (1578–1657), Descartes, Julien Offroy de La Mettrie (1709–1751), Thomas Hobbes (1588–1679), and Robert Whytt (1714–1766).

RENAISSANCE AND REFORMATION

The **Renaissance**, meaning "rebirth," originated in southern Italy in the 14th century, eventually spreading to Northern Europe. It promoted innovative developments in art, literature, architecture, music, mathematics, and—eventually—religion and science. Humanistic thinkers such as Francesco Petrarch (1304–1374), Giovanni Pico della Mirandola (1463–1494), and Desiderius Erasmus (c. 1466–1536) were highly critical of the institutional hierarchy and dogmatism of the established church and recommended a return to a more personal relationship with God. With greater access to classical literature, Renaissance humanists rediscovered the ancient Greek theorists and found much to admire in their focus upon the psychology of human life. They rediscovered Plato, who came to rival Aristotle as the classical authority, although Aristotle continued to be admired for his original works, as opposed to the sterile appropriations of his natural philosophy that had become fossilized as church dogma. Indeed, the Renaissance deserves to be characterized as a period of rediscovery as much as rebirth, since it was largely grounded in the recovery and retranslation of classical texts, which came to be admired for their intrinsic merits and celebration of humanity.

Petrarch is often treated as the founder of **Renaissance humanism**, insofar as his writings heralded the increased focus on the psychology of human individuals, including their place in the social and political order. Petrarch was critical of the sterility of scholastic thought and particularly the overly rigid Aristotelianism at the heart of Christian dogma. He celebrated the critical and naturalistic thought of the ancients and their focus on human capacities and potential. In religion, he recommended a return to the more personal and spiritual form of religion practiced by Augustine, presaging the later Protestant Reformation.

The Renaissance commitment to human potential was expressed in Pico's famous oration on the dignity of humanity, in which he located humankind as poised between the lower animals and the angels: capable of degenerating to bestiality, but also endowed with almost unlimited potential for creative intellectual, moral, and spiritual development. According to Pico, God had allowed humans to determine the limits of their own nature.

Somewhat paradoxically, in Renaissance humanism faith in the potential of humanity went hand in hand with skepticism about human pretensions to knowledge. Erasmus, in *The Praise of Folly* (1512), caricatured the dogmatic and superstitious beliefs of medieval scholasticism and contrasted the pretentious ceremony and hierarchy of the Church with the simple humility and humanity of Christ. Paracelsus rejected the classical authority of Galen and Avicenna in the realm of medicine, which he claimed should be founded upon empirical learning, although his own practice was heavily infused with astrology and mysticism.

The cautious skepticism of earlier humanists was eclipsed by the radical skepticism of Michel de Montaigne (1533–1592), who resurrected the arguments of Greek skeptics such as Pyrrho. He maintained that neither sense experience nor reason could yield knowledge of the natural and spiritual world. While few shared Montaigne's depth of skepticism, his advocacy of such an extreme position stimulated later defenses of autonomous rationality and the scientific method, notably those advanced by Descartes and Bacon.

The Renaissance promoted pioneering explorations of human nature in the art and anatomy of Leonardo da Vinci (1452–1519), the political writings of Niccolò Machiavelli (1469–1527), and the poetry and drama of William Shakespeare (1564–1616), but did little to advance the systematic scientific study of human psychology. However, it did witness the first attempts to apply medical and psychological theories to the development of education, most notably in the work of the Spanish humanist Juan Luis Vives (1492–1540). Born in Valencia, Vives was educated at the universities of Paris and Louvain, where he befriended Erasmus. His reputation as a teacher and scholar in the Netherlands later earned him a position at Oxford University (from 1523–1528), where he was supported by Thomas More and Henry VIII. After Henry's dispute with More over his divorce from Catherine of Aragon and More's subsequent execution, Vives returned to the Netherlands. There he completed *De Anima et Vita* (1538), in which he argued that knowledge of human physiology and psychology should be applied to the improvement of educational practice and the humane treatment of the insane.

Vives is best remembered for his comprehensive treatment of the associative principles of similarity, contrast, and contiguity (Brett, 1912–1921), which has led some to characterize him as the "father of modern psychology" (Clemens, 1967). Although his treatment of association followed Aristotle's general account, Vives tried to link the operation of memory to humoral physiology and cited many more examples of associationist principles than Aristotle. In many respects, Vives was a transitional figure, who retained great respect for classical authorities and deviated little from them in practice, but also conceived of the study of human psychology as a form of naturalistic knowledge grounded in observation:

> The study of man's soul exercises a most helpful influence on all kinds of knowledge. . . . This treatment of the development of knowledge within our souls will proceed parallel with the natural order.
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> —(*De Disciplinis*, cited in Clemens, 1967, p. 221)

Reformation

Dissatisfaction with the sterility, pomp, and hierarchy of the medieval Church eventually produced the religious movement known as the **Reformation**, spearheaded by Martin Luther (1483–1546), the Augustinian monk and professor at Wittenberg University who initiated the movement by nailing his 95 objections to the door of Wittenberg Cathedral in 1517. Luther's revolt was motivated by his objections to the Church's sale of indulgences (papal pardons for sins), a form of fund-raising promoted by the revolution in printing, which also enabled Luther's objections to be rapidly disseminated throughout Europe. Luther advocated a simpler and more spiritual approach to God and initially hoped for internal reform within the Church. However, he later rejected the philosophy of Aristotle and the authority of the pope, which led to his excommunication in 1521.

The form of Protestant religion originally developed by Luther, who emphasized individual faith, conscience, and attention to scripture in contrast to the hierarchical pomp and ritual of the established Church, represented an intellectual liberation of sorts. However, it very quickly rigidified into its own forms of institutionalized dogma as Protestantism spread throughout Europe. The ideal of individual conscience was converted into the ideal of conscience in obedience to scripture as interpreted by Luther and John Calvin (1509–1564), whose uncompromising doctrines about predestination exemplified a harsh and unforgiving attitude to sin. As the reformers attained positions of authority and power in Protestant states and provinces, they were at least as zealous in their persecution of heretics and dissenters as the traditional medieval Church.

One consequence of the Protestant Reformation was the institutional confirmation of the Aristotelian theories of Aquinas as the doctrinal foundation of

Roman Catholicism, as affirmed by the Council of Trent (1545–1563). It also seems to have encouraged a more vigorous and violent response to heretics, witches, and other dissenters by the Catholic Inquisition. The late 15th and 16th centuries marked the high point of religious repression in Europe, including the suppression of scientific works and the persecution of individual scientists.

Michael Servetus (c. 1511–1533), the Spanish anatomist who rejected Galen's account of the circulation of the blood in the heart and who was one of the first to identify pulmonary circulation, made the mistake of sending a copy of his "On the Restoration of Christianity" to John Calvin in Geneva. Calvin denounced Servetus to the Catholic Inquisition, and he was arrested and sentenced to death by burning. Servetus managed to escape, but he was later recaptured in Geneva and burned by Protestant reformers, while the Catholic Inquisition burned his effigy and his books. Although Servetus was persecuted for his religious rather than his medical views, the Reformation did little to promote the spirit of intellectual curiosity that motivated him.

THE SCIENTIFIC REVOLUTION

The period characterized as the scientific revolution was marked by revolutions in theory, particularly in astronomy, physics, and medicine. The most famous of these was the overthrow of the Aristotelian and Ptolemaic **geocentric** (Earth-centered) **theory** in favor of the Copernican **heliocentric** (sun-centered) **theory**. According to Ptolemaic theory, the fixed Earth is the center of the universe, with the sun and other planets revolving in circular orbits around it. Yet since the time of Aristotle, it had been known that planets do not move in perfect circular orbits. As observed from Earth, their orbits appear to be erratic, looping backward in their paths from time to time. To accommodate this "wandering," or retrograde, motion, Ptolemy had introduced a system of **epicycles**, or circles within circles, and this system had been modified and extended to a level of great complexity by later astronomers. The Ptolemaic theory served as an effective predictive and navigational device for centuries and was in accord with common sense. The planets appear in motion to the naked eye, and everyday experience seems to confirm a stationary Earth (we don't feel it moving, and don't fall off).

The Copernican Revolution

Nicholaus Copernicus was a Polish monk who studied at the universities of Cracow, Bologna, Ferrara, and Padua. He advanced his heliocentric theory in *On the Revolution of the Celestial Spheres,* published in 1543. In this work, Copernicus argued that the motions of the planets might be better explained by supposing that the sun, not Earth, is the fixed center of the universe and that Earth and other planets traverse circular orbits around it.

This was not a new hypothesis. Aristarchus (c. 310–c. 230 BCE) had first advanced it about 1,800 years earlier. Copernicus noted this and also ascribed an earlier version of his theory to the Pythagorean mathematician Philolaus (b. c. 480–480 BCE). Many were skeptical of the Ptolemaic system, since its complexity seemed inconsistent with Pythagorean requirements of simplicity and harmony. This was the view, for example, of Domenico Novara, professor of mathematics and astronomy at the University of Bologna, during the period Copernicus was in attendance.

Copernicus's heliocentric theory accommodated the same observational data as the developed geocentric theory. However, Copernicus was not able to do away with Ptolemy's system of epicycles, although he was able to reduce their number. Copernicus eliminated one serious anomaly of the geocentric theory, concerning the orbital times of planets. According to the geocentric theory, the moon, the closest planet to Earth, completes its orbit in four weeks, whereas the sun, which is furthest from Earth, takes only one day. According to the heliocentric theory, the orbital times of the planets vary inversely with their distance from the sun. Copernicus also provided an explanation of observed changes in planetary brightness, a problem for the geocentric theory but a natural consequence of the heliocentric theory (Dolling, Gianelli, & Statile, 2003). However, the Copernican theory had its own problems, notably the failure to detect **stellar parallax**, the variation in the angular separation of the stars, which was a crucial implication of Earth's projected orbit around the sun.

Copernicus's work was published posthumously, although his delay in publishing appears to have had more to do with his anticipation of the incredulity with which he thought his theory was likely to be received than out of any fear of persecution by the Church. Summaries of his conclusions had been circulated for years before, and one was published in 1540. It was not until Kepler's forceful advocacy of the Copernican theory in the late 1590s and the publication of Galileo's *Letters on the Solar Spots* (1613/1957) that the Copernican theory encountered serious opposition from the Church.

Realism and Instrumentalism The publication of *On the Revolution of the Celestial Spheres* initially encountered less opposition than might have been expected partly because the Lutheran theologian Andreas Osiander (1498–1552), who had been authorized to see Copernicus's work through the press after his death, added an introduction. Osiander suggested that Copernicus's work ought not to be read as a potentially true description of astronomical reality (of the relative positions and motions of the planets), but as a useful mathematical fiction that accommodated the planetary motions:

> For it is the job of the astronomer to use painstaking and skilled observation in gathering together the history of the celestial movements, and then—since he cannot by

> any line of reasoning reach the true causes of these movements—to think up or construct whatever causes or hypotheses he pleases such that, by the assumption of these causes, those same movements can be calculated from the principles of geometry for the past and for the future too . . . For it is not necessary that these hypotheses should be true, or even probably; but it is enough that they provide a calculus which fits the observations.
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> —(1543/2003, p. 43)

Osiander was an *instrumentalist*, a proponent of the view that scientific theories are merely calculative devices or "fictions" employed to predict observations or "save the appearances" and that the best theory is simply the most economical predictive device. Copernicus himself was almost certainly a *realist,* a proponent of the view that theories purport to describe reality and that the best theory is the one that provides the most accurate description of reality, as were later defenders of the Copernican system such as Galileo and Kepler. Kepler, who revealed the identity of Osiander as the author of the introduction to *On the Revolution of the Celestial Spheres* in the *New Astronomy* (1609), claimed that he founded astronomy on real causes and not fictional hypotheses.

Not all theologians shared Osiander's instrumentalist views. The Jesuit Christopher Clavius (1538–1612), who was also a realist, argued that Copernicus had simply saved the appearances by deducing true observational predictions from *false* theoretical assumptions. Clavius noted that there was nothing remarkable about this, since true conclusions (or predictions) can be deduced from any number of false assumptions. Thus, to take a modern example, the true conclusion "all metals are conductors" can be deduced from the true premises "all metals are elements with free electrons in their outer shells" and "all elements with free electrons in their outer shells are conductors" *and* from the false premises "all metals are elements containing electronic fluids" and "all elements containing electronic fluids are conductors." According to Clavius, the Copernican theory was simply false and inferior to the Ptolemaic theory, which he held to be consistent with both the principles of astronomy and Christian theology.

The Reception of the Copernican Theory These sorts of considerations led the Inquisition, under Cardinal Bellarmine (1542–1621), to adopt the view that the Copernican theory could be judged superior to the Ptolemaic theory only in terms of its economy as a mathematical model or calculation device and that to defend its physical truth was "formally heretical." *On the Revolution of the Celestial Spheres* was placed on the Catholic Index of Prohibited Books in 1616.

Galileo Galilei (1564–1642), who had indicated his support for the Copernican theory in *Letters on the Solar Spots* (1613/1957), was warned about the judgment of the Inquisition. For a few years he remained quiet, and the new Pope Urban VIII turned a blind eye to the unorthodox doctrines that Galileo advanced in *The Assayer* (1623/1957). However, in 1632 Galileo published *Dialogue Concerning the Two Chief World-Systems, Ptolemaic and Copernican.* This work was immediately prohibited, and the following year Galileo was imprisoned. Rheumatic and near blind at age 70, he was examined by the Inquisition and shown the instruments of torture. He was ordered to do penance for three years (while under house arrest) and to recant the Copernican doctrine. According to legend, at the end of his recantation he muttered under his breath "And yet it moves." The Catholic Church absolved him of his intellectual sins in 1992.

Others were not so lucky. It was bad enough that Copernicus had undermined the Aristotelian thesis, so congenial to Christian theology, that Earth and humankind are at the privileged center of creation, by suggesting that Earth is just one of the many planets orbiting the sun. The Dominican monk and astronomer Giordano Bruno (1548–1600) went one stage further and suggested that Earth is merely one (insignificant) planet among many in an infinite universe. In *On the Infinite Universe and Worlds* (1584/1950) Bruno declared that the debate between the Aristotelians and Copernicans about whether Earth or the sun is the center of the universe is vacuous, since "as the universe is infinite, no body can properly be said to be in the center of the universe or at the frontier thereof." In 1592, after many years of wandering Europe, Bruno unwisely let himself fall within the reach of the Inquisition. After seven years in prison, he was finally tried and condemned to death by burning at the stake (although it is unlikely that he was condemned to death for his scientific views—he also denied the Immaculate Conception and identified the pope with the Beast of Revelations).

Although the Copernican system eventually came to displace the Ptolemaic system, the "Copernican revolution" in astronomy was not an overnight affair. As late as 1669, the year Isaac Newton attained his professorship at Cambridge University, the Ptolemaic theory was still being defended, and opposition to the Copernican theory continued in France into the 18th century. At the time of Copernicus, there were no empirical grounds for preferring his system to that of Ptolemy, since both theoretical systems accommodated most of the available observational data.

Gradually fortunes shifted in favor of the Copernican theory. Johannes Kepler (1571–1630) was a dedicated Pythagorean who believed that God had created the world in accord with mathematical harmonies. He was employed as a research assistant to Tycho Brahe (1546–1601), the Danish astronomer, when the latter took up the position of royal mathematician at the court of the German King Rudolph II in Prague (where he was engaged in the preparation of military horoscopes). Working with Brahe's mass of accumulated observational data, which he inherited upon Brahe's death in 1601, Kepler eliminated many of the artificialities of the Copernican theory by supposing that the planets move in elliptical rather than circular orbits around the sun. In the *New Astronomy* (1609), he demonstrated that the orbital velocities of the planets vary with their distance from the sun, increasing as they approach the sun and decreasing as they move away from it.

In addition, after hearing of the invention of the telescope by Dutch lens crafters, Galileo immediately constructed his own and proceeded with record speed to observe the mountains and valleys of the moon, the moons of Jupiter, the phases of Venus, and the rings of Saturn, as well as a multitude of previously undetected stars. These observations undermined the general Aristotelian and Ptolemaic position. The mountains and valleys of the moon indicated that at least one celestial body is not perfectly spherical, and the observation of new stars indicated that the stars are at different distances from Earth and not fixed to a celestial sphere. The moons of Jupiter appeared to constitute a miniature Copernican system, since their orbits vary with their distance from Jupiter, and the phases of Venus could only be explained in terms of a sun-centered orbit.

None of this demonstrated the outright superiority of the Copernican theory, but it convinced many people. The crucial observation came with the telescopic observation of the stellar parallax predicted by the Copernican theory alone, although most astronomers had already abandoned the Ptolemaic theory by the time this was observed by Friedrich Bessel (1784–1846) in 1838, nearly 300 years after the publication of Copernicus's theory.

Galileo and the New Science

The scientific revolution was more than a revolution in astronomical—and physical and medical—theories: It amounted to a full-scale revolution in intellectual attitude. Prior to this time, many scholars were content to assess theoretical claims by reference to their consistency with classical and religious authorities. Famously, some of Galileo's colleagues at the University of Pisa refused to look through his telescope because they considered it redundant: They maintained that astronomical questions had already been settled by Aristotle and scripture. Yet from around the 16th century onward, natural philosophers came to adopt the view that theories ought not to be accepted until they have been empirically tested, ideally via what came to be known as a crucial experiment, enabling scientists to adjudicate between competing theoretical explanations of the same range of empirical data. They eventually became what the ancients and medievals had deplored, *empiriks*.

Galileo best epitomized this new empirical attitude. He was not prepared to accept or reject the Ptolemaic or Copernican theories on the basis of classical or religious authority and entered the astronomical debates only after he had developed the telescope and made what he believed to be crucial observations in support of the Copernican theory. Although earlier investigators anticipated him in both theory and practice, none matched his ability to integrate and propagate those elements that have since come to be treated as constitutive of modern science.

Galileo was appointed professor of mathematics at the University of Pisa at the age of 25. He served as professor of mathematics at the University of Padua from 1592 until 1610, when he became mathematician-in-residence to the Grand Duke of Tuscany. He made important contributions to astronomy and physics, subjecting entrenched Aristotelian theories to critical empirical scrutiny. He continued his scientific work right up to his death in 1642, albeit in secret, since his later years were spent under house arrest imposed by the Inquisition. His last work, *Dialogue Concerning Two New Sciences* (1638/1974), was smuggled out of Italy for publication.

Galileo was committed to the empirical evaluation of scientific theories and the development of instruments that enable and facilitate the testing of scientific theories. He constructed his first telescope in 1609 to test the Copernican theory. He also took a critical empirical look at Aristotle's theory of falling bodies, according to which bodies of different weight fall with different velocities. He demonstrated that bodies of different weight, such as a 100-pound cannon ball and 1-pound musket ball, fall with approximately the same velocity (according to legend, by dropping them off the Leaning Tower of Pisa). Using an improved water clock and a gently sloping inclined wooden plane down which he released polished bronze balls, Galileo developed and tested his own theory of falling bodies. This led him to recognize that forces act on bodies not to produce motion, as Aristotle had argued, but to change it, or produce acceleration.

Galileo not only rejected particular Aristotelian theories, but also the general form of Aristotelian explanation in physics. He renounced all attempts to explain the motion of bodies in terms of Aristotelian final causes, in terms of their propensity to move toward their "natural resting place," and employed only efficient causal explanations of matter in motion. This latter type of explanation, in terms of antecedent conditions sufficient to produce an effect, came to be characterized as **mechanistic explanation** and became associated with the popular 17th-century conception of the universe as a giant (usually clockwork) mechanism, governed by fixed laws of nature.

Galileo also insisted that the business of science is to explain quantitative and not merely qualitative changes and to do so by reference to quantitative changes in fundamental variables such as time, space, and motion, which led him to declare that mathematics is the language of science:

> Philosophy is written in this grand book, the universe, which stands continually open to our gaze. But the book cannot be understood unless one first learns to comprehend the language and read the letters in which it is composed. It is written in the language of mathematics.
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> —(1623/1957, pp. 237–238)

Galileo also reprised the ancient distinction between primary and secondary qualities. He maintained that primary qualities, such as size, shape, and motion, are real properties of material bodies and explain how bodies affect our senses. Secondary qualities, such as colors, tastes, and smells, are nothing more than the manner in which material bodies affect our senses:

> Hence I think that tastes, odors, colors and so on are no more than mere names so far as the object in which we place them is concerned, and that they reside only in the consciousness. Hence if the living creature were removed, all those qualities would be wiped away and annihilated.
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> —(1623/1957, p. 274)

Galileo explained differences in secondary qualities, such as tastes and smells, in terms of differences in primary qualities, such as the shapes, sizes, and velocities of microscopic particles:

> There are bodies which constantly dissolve into minute particles, some of which are heavier than air and descend, while others are lighter and rise up. The former may strike upon a certain part of our bodies that is much more sensitive than the skin, which does not feel the invasion of such subtle matter. This is the upper surface of the tongue; here the tiny particles are received, and mixing with and penetrating its moisture, they give rise to tastes, which are sweet or unsavory according to the various shapes, numbers and speeds of the particles.
>
> —(1623/1957, p. 276)

Galileo's new science represented an integration of the ancient naturalist (Ionian) and mathematical (Pythagorean) traditions. It also marked a new beginning, by combining these traditions with a new emphasis on empirical and experimental evaluation and the rejection of final causal explanation in favor of efficient causal or mechanistic explanation. These paradigmatic elements are also to be found in the work of the major scientists of the scientific revolution, such as Robert Boyle (1627–1691), Descartes, William Gilbert (1544–1603), William Harvey (1578–1657), Robert Hooke (1635–1703), Kepler, and Newton.

In these important respects, the work of such theorists was discontinuous with the work of most ancient and medieval theorists and marked a decisive break with the prior historical tradition. However, the scientific revolution was neither as sudden nor as revolutionary as its name might suggest. As noted earlier, anticipations of the new science can be found in the writings of scholastics such as Robert Grosseteste, Roger Bacon, and William of Occam; and empirical research played a significant role in the work of Aristotle, Alcmaeon, Hippocrates, and Galen. Although Galileo made much of his own break with the Aristotelian tradition, by his own day that tradition had become pretty eclectic. The doctrines that form the basis of Galileo's *Assayer*, for example, are to be found in his notes from his Aristotelian teachers at the Jesuit Collegío Romano (Wallace, 1984).

However, the rejection of classical orthodoxy came to play a major role in the rhetoric of the new science. Post-Galilean thinkers came to see themselves as making a new scientific beginning by breaking with tradition, rejecting ancient and medieval theories precisely because they were not empirically grounded. Thus, to take but one of many examples, Descartes felt obliged to preface his study of physiological psychology in *The Passions of the Soul* (1649) with the following remarks:

> The defects of the sciences we have from the ancients are nowhere more apparent than in their writings on the passions. . . . The teachings of the ancients about the passions are so meagre and for the most part so implausible that I cannot hope to approach the truth except by departing from the paths they have followed. This is why I shall be obliged to write just as if I were considering a topic that no one had dealt with before me.
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> —(1649/1985, p. 328)

Andreas Vesalius and the Scientific Revolution in Medicine

Galileo did not reject Aristotle's astronomical and physical theories out of hand, but only when they failed to survive empirical evaluation. Similarly, Andreas Vesalius (1514–1564) subjected the classical medical theories of Galen to empirical evaluation and found them wanting. A native of Belgium, Vesalius came from a line of royal physicians. He studied at the universities of Louvain and Paris and was appointed professor of surgery and anatomy at the University of Padua.

The dissection of cadavers had become commonplace in medical teaching by the time Vesalius took up his professorship. However, a butcher or barber would usually conduct the dissections. They would cut portions from a cadaver for a demonstrator to display to students, while the lecturer read from Latin translations of Galen or Avicenna.

Vesalius: dissection; cover plate of *On the Fabric of the Human Body* (1543).

Vesalius performed his own dissections and demonstrations and quickly identified many errors in Galen, which led him to conclude that much of Galen's system was based upon the physiology of pigs and goats rather than that of humans.

In 1543 Vesalius published his revolutionary *On the Structure of the Human Body*, which contained detailed descriptions and illustrations of the bones, muscles, veins, arteries, viscera, and brain of the human body. Although his challenges to Galen generated the same reactionary response as Galileo's challenges to Aristotle, his pioneering studies transformed medical theory and practice. He abandoned his own research in 1544 when he was appointed court physician to the Emperor Charles V, but his work was continued by his student Realdo Columbus (c. 1516–1559), who made important contributions to the study of circulation and respiration, and by later generations of anatomists such as Giovanni Battista Morgagni of Padua (1682–1771), Hermann Boerhaave (1668–1738), Joseph Lieutaud (1703–1780), and William Hunter (1718–1783).

Francis Bacon and the Inductive Method

One of the most articulate advocates of the new science was the Englishman Francis Bacon (1561–1626), who titled his major work on scientific method the *New Organon*, or "new method," (1620/1994), in explicit contrast to the Aristotelian corpus, which had come to be known as the *Organon*. Bacon was educated at the University of Cambridge, which he entered at age 13, and was admitted to the bar after studying at Gray's Inn. Although his attempts to obtain a high government position were thwarted (or at least ignored) by Queen Elizabeth I, his star rose (at least temporarily) when James I gained the throne in 1603. Bacon acquired various titles, including a knighthood, and high government office; he was appointed Attorney General in 1613 and Lord Chancellor in 1618. In 1621 he was publicly disgraced and imprisoned for having accepted bribes during his tenure as Lord Chancellor. In his defense, Bacon claimed that although he had taken bribes, he had not allowed them to influence his judgment. He spent his last years in seclusion and died as a consequence of one of his own experiments. He caught a cold while stuffing a chicken with snow in order to assess its utility as a preservative.

Bacon was a harsh critic of ancient and medieval natural philosophy and an optimistic and spirited promoter of the new science. He argued that the veneration of the ancients and the contemplative ideals of scholastic thought were major obstacles to the progress of scientific knowledge. He recommended a more active and critical approach, in which "vexed nature" was interrogated through experimental intervention. He maintained that a true science of nature should be grounded in mechanical crafts such as alchemy rather than scholastic contemplation and reflection (although he was also critical of many alchemical practices, and practitioners such as Paracelsus).

Bacon was a committed realist and materialist, who believed that the "secrets of nature" could be revealed through observation and experiment. He was dismissive of Renaissance skepticism, which he thought could be overcome by the employment of a proper method for revealing the "subtlety of Nature." He followed Galileo in maintaining that final causality has no place in the explanation of the motion of physical bodies, although, like many other 16th- and 17th-century scientists (including Galileo), he acknowledged that final causal explanations are legitimate in their appropriate psychological domain:

> The final cause, so far from assisting the sciences, actually corrupts them, except for those concerned with human actions.
>
> —(1620/1994, p. 134)

Bacon is often treated as a champion of the inductive method, who abjured hypothetical speculation in favor of careful and systematic observation. He did claim that his own method involved cautious inferential ascent from carefully established "natural and experimental histories" to the establishment of theoretical axioms:

> But there will be hope for the sciences when, and only when, ascent is made by the right kind of ladder, through an uninterrupted, connected series of steps, from particulars to lesser axioms, one above the other, and last of all to the most general.
>
> —(1620/1994, p. 110)

However, Bacon was dismissive of the Aristotelian method of enumerative induction, through which general truths of nature are derived by generalization from the observation of positive instances of correlation (by generalizing that "All swans are white" on the basis of the observation of a number of white swans, for example). He insisted that true natural science should be based upon eliminative induction, in which causal conditions are identified via the falsification of alternative causal hypotheses:

> For induction that proceeds through simple enumeration is childish, its conclusions are precarious, and open to danger from a contradicting instance, and it generally makes its pronouncement on too few things, and on those only that are ready to hand. But induction that will be of use for the discovery and demonstration of the arts and sciences must analyse Nature by proper rejections and exclusions, and then, after a number of negatives, come to a conclusion on the affirmative instances.
>
> —(1620/1994, p. 111)

Bacon did not advocate a narrow empiricist science restricted to observational correlation. He was well aware of the need for creative invention in the formulation of hypotheses and championed the role of novel prediction in empirical evaluation. He claimed that hypotheses should be evaluated by their utility in "discovering new works": "axioms properly and methodologically applied can very well point to and indicate new particulars" (1620/1994, p. 50). He recognized that different hypotheses can provide formally adequate explanations of the same empirical data (as demonstrated by the Copernican debates) and recommended that such conflicts between hypotheses be adjudicated by a crucial instance (1620/1994, p. 210): a prediction affirmed by one hypothesis but denied by the other (such as the different predictions about whether light accelerates or decelerates in moving from a less dense to a denser medium offered by the particle and wave theories of light). Bacon characterized such a prediction as an **Instance of the Fingerpost**, which serves as a "pointer" to the correct theory (in England, signposts at rural crossroads often have wooden fingers pointing in the direction of nearby villages).

Bacon was opposed to the Aristotelian rational intuition of causal principles and essential forms, of proceeding directly to very general theories or axioms:

> The understanding must not be allowed to leap and fly from particulars to remote and nearly the most general axioms . . . and from their [supposed] unshakeable truth, to prove and deliver intermediate axioms.
>
> —(1620/1994, p. 110)

However, as Newton was shortly to demonstrate with his theory of gravitation, a speculative "leap" to very general principles could also be enormously productive.

Nevertheless, Bacon's account of inductive ascent from observations to increasingly general hypotheses and theories does describe the practice of many of the contributors to the scientific revolution. Robert Hooke and Robert Boyle derived their laws of elasticity and gas expansion from tables of correlation, and the works of William Gilbert on magnetism (*On Magnetism*, 1600/1958), William Harvey on the circulation of the blood (*On the Circulation of the Blood,* 1628/1989) and Isaac Newton on optics (*Opticks*, 1704/1952) include many careful descriptions of observed effects and tentative empirical laws, followed by conclusions that develop cautious speculative theories to integrate and explain these tentative laws.

Bacon was one of the first theorists to stress that scientific knowledge enables scientists to predict and control the natural world, or establish "dominion over nature." He famously claimed that

> Human knowledge and human power come to the same thing, for where the cause is not known the effect cannot be produced. We can only command Nature by obeying her, and what in contemplation represents the cause, in operation stands as the rule.
>
> —(1620/1994, p. 43)

He was greatly impressed by the technological potential of scientific discoveries, citing the recent inventions of printing, gunpowder, and the compass:

> It is worth noticing the great power and value and consequences of discoveries, in none more obvious than those three which were unknown to the ancients . . . namely, the arts of printing, gunpowder, and the compass. For these three have changed the whole face and condition of things throughout the world, in literature, in warfare and in navigation.
>
> —(1620/1994, pp. 130–131)

In claiming that scientific theories should be judged by the success of their "works," Bacon also stressed their contribution to the human condition. In the *New Atlantis* (1627/1966), he envisioned a developed inductive science capable of relieving human pain, curing disease, and extending the life span.

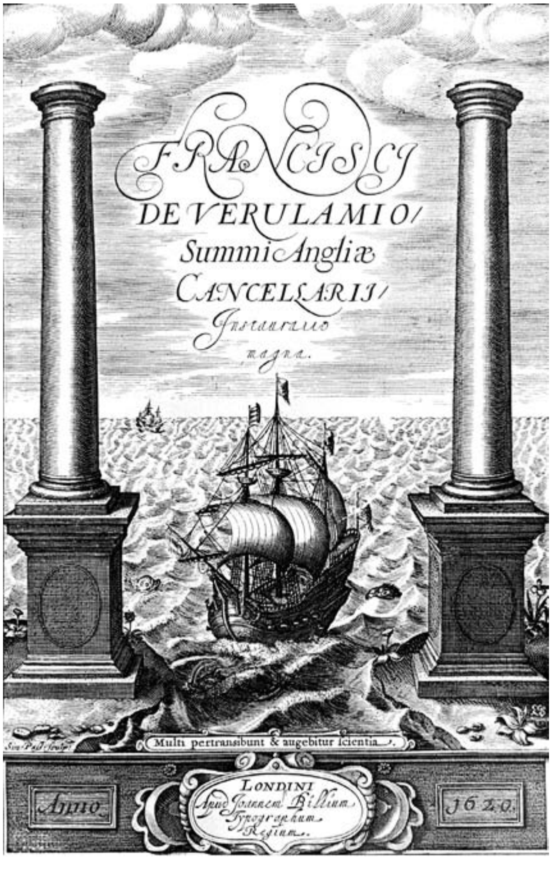
Like many other advocates of the new science, Bacon recommended that practitioners should abandon the theories of the ancients and medievals and begin anew:

> We can look in vain for advancement in scientific knowledge from the superinducing and grafting of new things on old. A fresh start must be made, beginning from the very foundations.
>
> —(1620/1954, p. 51)

In representing the new science as setting the course for new scientific discoveries, Bacon likened himself to Columbus setting out to discover the Americas:

> And so my conjectures, which make what is hoped for probable, are set out and made known; just as Columbus did, before his wonderful voyage across the Atlantic Ocean.
>
> —(1620/1954, p. 103)

The frontispiece of the *New Organon* depicted a ship setting out on uncharted waters.



Frontispiece of Francis Bacon's *Novum Organum* (1620). Ship of Knowledge setting out on uncharted waters.

Social Dimensions of Science Bacon was one of the earliest practitioners of the psychology and sociology of science and documented a variety of cognitive deficits and social biases, which he called "idols which beset men's minds." He argued that the method of inductive ascent was the best means of surmounting these cognitive and social dimensions of human nature.

Bacon characterized as **Idols of the tribe** those innate human propensities to project a greater degree of regularity in nature than can actually be found, to presume that the "subtleties" of nature can be understood through familiar analogies, and to adhere to favored theories in the face of empirical falsification. The net result of such cognitive deficits was what Bacon called "wishful science." Bacon characterized as **Idols of the cave** those idiosyncratic products of individual human development that incline some men to fixate on novelty and others to overproject similarity or difference in nature. He cautioned scientists to be especially suspicious of any theoretical notion about which they were individually enthusiastic.

Bacon characterized as **Idols of the marketplace** those notions derived from common linguistic usage employed in the theoretical description of nature that impede the development of proper scientific terminology. He characterized as **Idols of the theatre** those theoretical systems that are socially maintained by the various schools of philosophy as received dogma (such as the Ptolemaic theory) and form the bases of most forms of education.

Although he identified some of the social dimensions of human nature that bias scientific thought, Bacon was also a forceful advocate of the social community of science. He recognized the benefits for scientific communication derived from the invention of printing and those that could be accrued through the social cooperation of scientists. In the *New Atlantis* (1627/1966) he envisioned a future society of scientists and technologists devoted to knowledge and discovery and petitioned King James I to finance the creation of cooperative research projects. Although personally unsuccessful in securing this goal in his own lifetime, the "Royal Society of London for Improving Natural Knowledge" was founded in London by Charles II in 1662 and implemented both Bacon's general vision and a number of his specific research projects.

Similar societies were founded in Europe at around the same time. The Academia del Cimento (Academy of Experiments) was founded in Florence in 1657, the Académie des Sciences in Paris in 1666, the Berlin Academy in 1700, and the St. Petersburg Academy in 1724 (Pyenson & Sheets-Pyenson, 1999). One consequence of the formation of these scientific societies was the development of the logic and practice of what came to be known as the experimental report. In the early meetings of the Royal Society, when the numbers were relatively small, members used to demonstrate their "effects" in front of their colleagues. When time constraints and the rapidly increasing membership made it impractical for most members to do so, they developed a convention that members should report their results by writing a "recipe" that would enable any other member to reproduce their effects. In this fashion the logic of experimental replication was born. These "recipes" were collected annually and published as the *Philosophical Transactions of the Royal Society* (Bazerman, 1988). Experimental reports in psychology, with their methods, design, and procedure sections, are direct descendants of these "recipes."

Philosophical Transactions, first published in 1665, became the model for later scientific journals, such as the German *Miscellanea Curiosa*, first published in 1670, and the French *Histoires et Mémoires*, first published in 1702. International scientific communication was also greatly enhanced by the emergence of scientific correspondents; these initial efforts developed into the institution of corresponding members of scientific societies. Henry Oldenburg (c. 1618–1647), the first secretary of the Royal Society, maintained an extensive correspondence network with members of the European scientific community, as did Marin Mersenne (1588–1648) in Paris.

The Newtonian Synthesis

In the conclusion of *On Magnetism* (1600/1958), William Gilbert had speculated that the planets are held in their orbits (and their matter held in cohesion) by a force analogous if not identical to magnetism. Isaac Newton (1642–1727) developed this speculation into the theory of universal gravitation. Born in Lincolnshire, Newton was educated at Trinity College, Cambridge, where he received his degree in 1665. In the two years following, he secluded himself in Lincolnshire to avoid the plague. This was perhaps Newton's most creative period: He developed the binomial theorem, invented the "method of fluxions" (calculus), and created the first reflecting telescope. It was also during this period that Newton first began to develop his theory of universal gravitation. He was appointed professor of mathematics at Cambridge in 1669. He became a fellow of the Royal Society in 1672 and was elected president in 1703. He published *Mathematical Principles of Natural Philosophy* (*Principia*) in 1687 and *Opticks* in 1704.

In 1696 Newton was appointed warden of the Royal Mint, in order that he might apply his mathematical talents to the reformation of the currency—although Voltaire maintained that he was appointed because the Treasurer, Lord Halifax, was besotted with Newton's niece. Throughout his life Newton engaged in running feuds with Robert Hooke and Gottfried Leibniz (1646–1716) over credit for the initial development of the "rectilinear inertial principle" and calculus. With respect to the development of calculus, an investigative committee of the Royal Society found in favor of Newton, but this was scarcely surprising, since Newton, in his capacity as president of the society, appointed the committee and authored its final report.

Newton's theory of universal gravitation was held to be a triumph of mechanistic explanation, since it integrated the laws of terrestrial and celestial motion propounded by Galileo and Kepler (or at least approximations of them) and successfully explained a wide range of empirical data, such as the motion of the tides and centrifugal motion. Newton also followed Galileo in assuming that quantified mechanistic laws could be extended to the atomistic components of material bodies, or **corpuscles** as Robert Boyle called them, and developed his own **corpuscularian theory of light** in *Opticks* (1704/1952), in which he treated light as a stream of material corpuscles. The triumph of the new mechanistic and mathematical science, based upon quantified efficient causal explanations of matter in motion, appeared complete.

Yet not everyone rushed to embrace Newton's gravitational theory, at least initially. It took almost 80 years for Newton's theory to displace Descartes' rival vortex theory on the continent of Europe. One of the advantages of Descartes' theory was that it explained why all the planetary orbits are in the same direction, which Newton's theory did not. However, Newton's theory eventually came to establish its supremacy, and deservedly so, since later Newtonians transformed what initially appeared to be empirical anomalies into substantive developments of Newtonian theory. For example, U. J. J. Leverrier (1811–1877) accommodated the initial failure of Newton's theory to correctly predict the orbit of Uranus by postulating another planet beyond Uranus, which led to the discovery of the planet Neptune.

Newton, who was a great admirer of Bacon, avowed that he had followed the method of inductive ascent in the development of his theories:

> Particular propositions are inferred from the phenomena, and afterwards rendered general by induction. Thus it was that the impenetrability, the mobility, and the impulsive force of bodies, and the laws of motion and gravitation, were discovered.
>
> —(1687/1969, p. 547)

Yet this was patently not the case with respect to the development of Newton's laws of motion, which involved the direct postulation of very abstract axioms. Newton's first law of motion states, "Every body continues in a state of rest, or of uniform motion in a right line, unless it is compelled to change that state by forces impressed upon it." Since no body actually moves in a right line, because every body is subjected to external forces, this cannot be established by induction (Losee, 1980).

MAN THE MACHINE

The mechanistic forms of efficient causal explanation that displaced teleological or final causal explanation in astronomy and physics were eventually extended to biology and psychology. One of the first and most influential mechanistic explanations of a biological process was William Harvey's account of the circulation of the blood (1628/1989), in which he claimed that the veins and arteries form closed loops through which the heart pumps blood.

The Spanish physician Gómez Pereira (1500–c.1559) extended mechanistic explanation to the whole human body. Pereira studied at the University of Salamanca, where he became acquainted with the work of the Merton mathematicians, notably Richard Swineshead's text on motion, *Liber Calculationum* (Bandrés & Llavona, 1992). In *Antoniana Margarita* (1554), Pereira employed Swineshead's theoretical system to explain the "vital" functions of animals in purely mechanistic terms. He explained all forms of animal behavior in terms of instincts and learned habits, without any reference to consciousness or reason, which he denied animals possessed. He provided a detailed account of reflexive behavior, in which he described how mechanical activation in sensory organs is transmitted by the nerves to the brain, which in turn activates nerves that produce mechanical movements in muscles. Pereira's account anticipated the theory of reflexive behavior later developed by Descartes.

René Descartes: Mind and Mechanism

René Descartes was born in La Haye, France, in 1596, and educated at the Jesuit College at La Flèche. He attained a degree in law from the University of Poitiers, but did not practice, since his share of the family fortune furnished him with independent financial means. He enlisted privately in the Dutch army in 1618 and, while serving at Ulm, had a dream "in a stove-heated room" that stimulated his interest in science and methodology. He traveled widely in Europe, returning to take up residence in Holland in 1628. During the next 20 years he changed his residence as many times, his whereabouts known only to his close friend in Paris, Marin Mersenne, with whom he corresponded but rarely saw. The reasons for his voluntary solitude are unclear, since few details of his private life are known. He never married, although he did have an illegitimate daughter, Francine, who died at the age of 5 in 1640.

Between 1629 and 1633 Descartes produced his major work on physics and mathematics, *The World*, but suppressed its publication when he heard of Galileo's troubles with the Inquisition. To no avail, as it turned out: Descartes' works were placed on the Catholic Index of Prohibited Books, as was *The World* when it was published posthumously in 1664. In this work Descartes presented his vortex theory of celestial motion, which dominated continental Europe in the late 17th and early 18th century, until it was eventually displaced by Newton's gravitational theory. Descartes introduced analytic geometry, with its system of what are now known as Cartesian coordinates, in *Discourse on Method* in 1637. His other major works on knowledge and the relation between mind and body were *Meditations on First Philosophy* (1641) and *The Passions of the Soul* (1649).

In 1649 Queen Kristina of Sweden (1626–1689) invited the now famous Descartes to be her personal philosopher in residence. He accepted, but it proved to be a fatal error. The philosopher who had developed many of his ideas in his bed (he created analytic geometry by meditating on a means of plotting the position of a fly on the roof above his bed) did not take kindly to Queen Kristina's tutorial schedule, which began at five in the morning, or the severe Swedish winter. He died of pneumonia within six months and was buried in a Swedish cemetery. His last words are reputed to have been "So, my soul, it is time to part."

Insult followed injury. In 1666 the French resolved to have Descartes' remains returned to his native land. The French ambassador to Sweden arranged to have the body exhumed and returned to France in a special copper coffin constructed for this purpose, but on exhumation it was discovered that the coffin was too short. The ambassador ordered that the head be severed from the body, to be returned to France separately. The body was shipped back to Paris, where it was buried in the church of Sainte-Genevieve-du-Mont, minus the right forefinger, which the ambassador had cut off as a souvenir. Unfortunately, the head did not make it back as quickly. It was purloined by a Swedish army captain and changed hands many times among private collectors of exotica before being returned to Paris in 1806. For many years it was shelved in the Musée de l'Homme, part of the National Academy of Sciences, where it remained until very recently (Boakes, 1984). The present curator was not happy that the head was being displayed among a collection of criminals and primitives and removed it from the shelf.

Descartes' skull.

According to the last report,[1] it is now housed in a drawer of one of his filing cabinets!

Descartes' Science Like Bacon, Descartes aimed to reconstruct human knowledge and dismissed the "shaky foundations" upon which the ancient and medieval tradition was based. He resolved

> never to accept anything as true if I did not have evident knowledge of its truth . . . to avoid precipitous conclusions and preconceptions, and to include nothing more in my judgments than what is presented to my mind so clearly and so distinctly that I had no occasion to doubt it.
>
> —(1637/1985, p. 120)

Although he ended up affirming many doctrines that were congenial to the Catholic Church, such as the existence of God and the immortality of the soul, Descartes insisted that any form of knowledge worth its name ought to be independently demonstrable through reason or empirical evidence.

Also like Bacon, Descartes affirmed the potential of the new mechanistic science to extend the power of humans over nature and to improve the human condition, in contrast to the contemplative natural philosophy of the scholastics. As he put it, the new science "opened my eyes to the possibility of gaining knowledge which would be very useful in life, and of discovering a practical philosophy which might replace the speculative philosophy taught in the schools":

> Through this philosophy we could know the power and action of fire, water, air, the stars, the heavens and all the other bodies in the environment . . . and we could use this knowledge . . . for all the purposes for which it is appropriate, and thus make ourselves, as it were, the lords and masters of nature. This is desirable not only for the invention of innumerable devices which would facilitate our enjoyment of the fruits of the earth and all the goods we find there, but also, more importantly, for the maintenance of health, which is undoubtedly the chief good and the foundation of all the other goods in this life.
>
> —(1637/1985, pp. 142–143)

Like Galileo, Descartes was committed to the primary and secondary quality distinction and abjured final causal explanation in physics. Indeed, in one fundamental respect his physics represented more of a paradigm of mechanistic explanation than Newton's physics, since Descartes conceived of motion as the

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[1] I owe this piece of information to a former graduate student, Mark Sheehan, who visited the Musée de l'Homme to view Descartes' skull.

rearrangement of bodies in space. According to Descartes' **vortex theory**, planets are held in their orbits by swirling vortices of "subtle matter," analogous to the motion of corks caught up in a whirlpool. Many held that such explanations in terms of "action-by-contact" were superior to explanations in terms of "action-at-a-distance," such as explanations postulating "occult" forces of gravitational or magnetic attraction, which seemed as dubious as Aristotelian final causal explanations in terms of bodies trying to reach their natural resting place.

In contrast to Bacon and Galileo, Descartes was a rationalist. He claimed that knowledge of the fundamental nature of material bodies could be attained only by rational intuition, since it is not given to us in the flux of sense experience. Thus Descartes maintained that some general theoretical principles are known a priori, independently of sense experience. For example, in his discussion of the melting of a piece of wax in the *Meditations* (1641/1985), he argued that we determine that extension (in space) is the essential property of material bodies through rational intuition rather than by sense experience, since we recognize that it is the only property that remains constant throughout changes in the perceived taste, smell, color, shape, and size of the wax (1641/1985, p. 20–21).

Descartes also claimed that very general principles of physics, such as "all motion is caused by impact or pressure" and "all bodies at rest remain at rest, and bodies in motion remain in motion, unless acted upon by some other body" (Newton's first law), could be rationally intuited, or deduced from rationally intuited principles (Buchdahl, 1969). He maintained that we have innate ideas and knowledge: that our ideas of God, infinity, and perfection, and our knowledge of the axioms of geometry and logic are so "clear and distinct" that they must be accepted as true, even though they may have no counterparts in our sense experience.

Descartes' ideal of knowledge was a deductive structure with rationally intuited axioms at its apex. His goal was to identify axioms, or "first principles," that were so certain that they were immune from error or doubt. Although he rejected Bacon's claim that such axioms must be established via inductive ascent, Descartes recognized that lower-level principles and laws have to be established by observation and experiment. For Descartes, rationally induced general laws place constraints on our theories of the motion of material bodies, but the particular content of laws governing their motion has to be empirically determined (Clark, 1982). Descartes' own work in optics and biology was based upon observation and experiment, and in the last chapter of *Discourse on Method* he acknowledged that competing scientific explanations can be adjudicated only by critical observations, or what Bacon called crucial instances:

> I know of no other means to discover this than by seeking further observations whose outcomes vary according to which of the ways provide the correct explanation.
>
> —(1637/1985, p. 144)

Animal Automatism One of Descartes' most significant contributions to the history of science and psychology was his application of the mechanistic principles of efficient causal explanation to the behavior of organic beings. In his *Treatise on Man* (the second part of *The World*), he advanced mechanistic explanations (in terms of matter in motion) of the biological and psychological functions of animals and humans. He maintained that

> the digestion of food, the beating of the heart and arteries, the nourishment and growth of the limbs, respiration, waking and sleeping, the reception by the external sense organs of light, sounds, smells, tastes, heat and other such qualities, the imprinting of the ideas of these qualities in the organ of the "common" sense and the imagination, the retention or stamping of these ideas in the memory, the internal movements of the appetites and passions, and finally the external movement of all the limbs . . . follow from the mere arrangement of the machine's organs every bit as naturally as the movements of a clock or other automaton [moving machine] follow from the arrangement of its counter-weights and wheels.
>
> —(1664/1985, p. 108)

One of Descartes' best known contributions in this area was his detailed description of **reflexive behavior**. Although Galen had identified simple reflexes such as the pupillary reflex, Descartes was the first to provide a detailed physiological account of reflexive behavior, which he characterized as automatic and involuntary:

> If someone suddenly thrusts his hand in front of our eyes as if to strike us, then even if we know he is our friend, that he is doing this only in fun, and that he will take care not to harm us, we still find it difficult to prevent ourselves from closing our eyes. This shows that it is not through the mediation of our soul that they close, since this action is contrary to our volition, which is the only, or at least the principle, activity of the soul. They close rather because the mechanism of our body is so composed that the movement of the hand towards our eyes produces another movement in our brain, which directs the animal spirits into our muscles that make our eyelids drop.
>
> —(1649/1985, pp. 333–334)

Descartes called such behavior reflexive because he believed that in the case of automatic and involuntary behavior, animal spirits are "reflected" in the brain in the fashion that light is reflected on the surface of a liquid (Boakes, 1984).

Descartes claimed that sensory organs are connected to pores in the brain via a system of "delicate threads" within the nerves and that the pores in the brain are capable of directing animal spirits though the nerves to the muscles. In the case of a person who withdraws a foot when it comes into contact with fire,

Man reflexively withdrawing foot from fire, illustrating nerve pathway to brain. From Descartes: *Treatise on Man* (1664).

Descartes supposed that the moving particles of the fire interact with receptors in the foot, which pull on the nerve threads connected to the pores of the brain. This action in turn causes the release of animal spirits, which flow through the nerves to the muscles of the foot, causing it to be withdrawn from the fire:

> Next, to understand how the external objects which strike the sense organs can prompt this machine to move its limbs in numerous different ways, you should consider that the tiny fibres (which, as I have already told you, come from the innermost region of its brain and compose the marrow of the nerves) are so arranged in each part of the machine that serves as the organ of some sense that they can be easily moved by the objects of that sense. And when they are moved, with however little force, they simultaneously pull the parts of the brain from which they come, and thereby open the entrances to certain pores in the internal surface of the brain. Through these pores the animal spirits in the cavities of the brain immediately begin to make their way back into the nerves and so to the muscles which serve to cause movements in the machine.
>
> —(1664/1985, p. 101)

Descartes claimed that this mechanistic reflexive form of explanation could be extended to all animal and much of human behavior and suggested that

> This will not seem at all strange to those who know how many kinds of automatons, or moving machines, the skill of man can construct with the use of very few parts, in comparison with the great multitude of bones, muscles, nerves, arteries, veins, and all the other parts that are in the body of any animal. For they will regard this body as a machine which, having been made by the hands of God, is incomparably better ordered than any machine that can be devised by man, and contains in itself movements more wonderful than those in any such machine.
>
> —(1637/1985, p. 139)

Descartes' conception of the living body as an **automaton** or "moving machine" may have been inspired by the mechanical statues found in the royal gardens of his day, such as those in the chateau of Saint-Germain-en-Laye outside Paris (which Descartes may have visited), powered by water and triggered by mechanical plates embedded in footpaths. The general form of Descartes' account did not mark much of an advance over the medieval theory of the inner senses. He retained Aristotle's common sense and Galen's "animal spirits," and many of the details of his account were empirically falsified within his own lifetime. However, Descartes' account was revolutionary because he applied mechanistic reflexive explanation not only to innate reflexes such as the pupillary reflex and involuntary behavior such as digestion, yawning, and sleeping, but also to many forms of animal and human behavior based upon learning and memory: "movements which are so appropriate not only to the actions of objects presented to our senses, but also to the passions and the impressions found in memory" (1664/1985, p. 108). Like 20th-century behaviorist psychologists, he maintained that the learned behavior of animals, and much of the learned behavior of humans, is as automatic and involuntary as innate reflexes and instincts and can be explained *without reference to consciousness or cognition*. Like Gómez Pereira, Descartes held that animals lack consciousness and reason, which he believed justified his own practice of dissecting live animals, whose yelps and howls he treated as merely mechanical noises.

Mind and Body Although Descartes believed that mechanistic reflexive forms of explanation could account for all animal behavior and some human behavior, he denied that they could account for voluntary human behavior. Descartes did not simply maintain, as many contemporary cognitive psychologists would maintain, that some human behaviors are nonreflexive because they involve some form of internal cognitive processing and thus require a more complex mechanistic explanation. Rather, he claimed that voluntary human behavior could not be explained mechanistically at all. According to Descartes, some human behavior is generated through the action of a distinct immaterial soul, whose essence is thought. Descartes was perhaps the most famous substance dualist and **interactionist**. He claimed that the immaterial mind, the seat of reason, consciousness, and will, interacts with the material body via the pineal gland in the brain, which enables the immaterial mind to direct the animal spirits to different muscles and generate different forms of behavior at will.

Why did Descartes hold such a view? It is easy to understand how he might have been motivated to do so. To extend mechanistic explanation to the human mind would have been to deny the existence of the immortal soul, still a dangerous heresy in Descartes' day. He was well aware of the fate of Bruno and Galileo and withdrew his general mechanistic work *The World* when he learned of Galileo's condemnation by the Inquisition. Later critics have speculated that Descartes did not really believe that human psychology is exempt from mechanistic explanation, but only publicly advocated such a view to avoid persecution (Lafleur, 1956). Julien Offroy de La Mettrie (1709–1751), who did extend the principles of mechanistic explanation to human thought and voluntary behavior, was one of the first to accuse Descartes of being a closet materialist about the mind.

Descartes' primary argument for his ontological distinction between mind and body was epistemological in nature and was part of his general project to set knowledge upon firm and certain foundations. In reaction to ancient and Renaissance skepticism about beliefs derived from our sense experience of the world, Descartes sought a "first principle" for his knowledge system that was so certain that it was immune from error or doubt. He followed Augustine and Avicenna in claiming that although he could doubt that he had a material body, he could not doubt that he existed as a thinking being, since thinking presupposes existence and doubting is a form of thinking. Consequently,

> observing, that this truth "I am thinking, therefore I exist" was so firm and sure that all the most extravagant suppositions of the skeptics were incapable of shaking it, I decided that I could accept it without scruple as the first principle of the philosophy I was seeking.
>
> —(1637/1985, p. 127)

Given that he could without contradiction or absurdity doubt the existence of his body (however exaggerated this doubt might be, including the imagination of an "evil demon" intent on deceiving him), but could not doubt his existence as a thinking being, Descartes claimed that he could not be identical to his body, since "Otherwise, if I had doubts about my body, I would also have doubts about myself, and I cannot have doubts about that" (1643/1985, p. 412).

Whatever the merits of this argument, Descartes' interactionist account of the relation between mind and body created a serious problem. How could an immaterial mind, with no physical properties or spatial location, interact with a material body extended in space? This was an especially pressing problem for Descartes, given his commitment to the efficient causal explanation of the motion of material bodies in terms of action by contact and his recognition of the intimate connection between mind and body, particularly in relation to the appetites and emotions:

> Nature also teaches me, by these sensations of pain, hunger, thirst and so on, that I am not merely present in my body as a sailor is present in a ship, but that I am very closely joined and, as it were, intermingled with it, so that I and the body form a unit.
>
> —(1641/1985, p. 56)

This problem, which had vexed Queen Kristina of Sweden, was one that Descartes never resolved.

Materialist critics such as Pierre Gassendi (1592–1655), Hobbes, and La Mettrie maintained that the functions of the human mind, including language and reasoning, could be ascribed to the brain and extended mechanistic causal explanation to encompass all human thought and behavior. Other critics defended mind-body dualism but rejected interactionism. Arnold Geulincx (1625–1669) and Nicholas Malebranche (1638–1715) held that God directly causes the regular correlation between mental and bodily states, a view known as **occasionalism**, and Leibniz claimed that God maintains the regular correlation of mental and bodily states through a **pre-established harmony** between mental and bodily states. However, neither position proved popular with later dualists.

Machine and Animal Intelligence Descartes offered arguments in support of his claim that mechanistic reflexive explanation could not be extended to voluntary human behavior that were independent of his epistemological arguments for mind-body dualism. He claimed that voluntary human behavior could always be distinguished from the behavior of animals or machines, even if such machines were physically modeled upon real people. These arguments are especially interesting because they anticipate late-20th-century debates about whether machines such as digital computers are capable of simulating language comprehension and problem solving.

According to Descartes, all machines, including animal automata, are incapable of language. Although suitable machines could be created (and animals such as parrots taught) to produce appropriate noises in appropriate stimulus situations—for example, to utter "I am in pain" when their receptors were damaged—Descartes claimed that

> it is not conceivable that such a machine should produce different arrangements of words so as to give an appropriately meaningful answer to whatever is said in its presence, as even the dullest of men can do.
>
> —(1637/1985, p. 140)

Although he acknowledged that machines could perform some complex tasks better than humans (a mechanical clock can measure time better than a person can), Descartes claimed that no machine is capable of problem solving in the form of rational adaptation to novel situations:

> Even though such machines might do some things as well as we do them, or perhaps even better, they would inevitably fail in others, which would reveal that they were acting not through understanding but only from the disposition of their organs.
>
> —(1637/1985, p. 140)

These arguments highlight the peculiar nature of Descartes' contribution to psychology. In claiming that mechanistic explanation could be extended to the realm of animal and human behavior, he emphasized the continuity of animal and human behavior with other material processes in nature. In denying that mechanistic reflexive explanation could be extended to human thought and voluntary behavior, he postulated a fundamental discontinuity between animals and human beings.

Descartes believed that the extension of mechanistic explanation to human thought and voluntary behavior undermined human freedom and claimed that the idea that humans are no different from animals posed a serious threat to morality and religion:

> For after the errors of those who deny God . . . there is none that leads weak minds further from the path of virtue than that of imagining that the souls of beasts are of the same nature as ours, and hence that after this present life we have nothing to fear or to hope for, any more than flies and ants.
>
> —(1637/1985, p. 141)

It is important to recognize that Descartes' arguments against animal and machine language and problem solving were independent of his arguments for mind-body dualism. Conwy Lloyd Morgan, the comparative psychologist, and John B. Watson, the behaviorist psychologist, were later materialists who also maintained that only humans have the capacity for language, and the question of whether machines such as digital computers are capable of genuinely creative problem solving remains a lively issue for contemporary psychologists and philosophers (Boden, 2003; Dreyfus, 1992).

Vitalism Descartes extended the principles of mechanistic explanation to the functions of the Aristotelian nutritive and sensitive souls, but not to those of the rational soul. In so doing, he took the revolutionary step of separating the principles of life and mind.

From the time of the ancient Greeks, the psyche had been treated as the actualizing principle of both life and mind. By maintaining that vital processes such as digestion, respiration, and sensory-motor reflexes are a product of the organized matter of the body machine, Descartes denied that the rational soul or mind is responsible for the life of the material body:

> In order to explain these functions, then, it is not necessary to conceive of this machine as having any vegetative or sensitive soul or other principle of movement and life.
>
> —(*Treatise on Man,* 1664/1985, p. 108)

This was the fundamental idea behind the mechanistic conception of biological functions. Or, as Descartes put it,

> And let us recognize that the difference between the body of a living man and that of a dead man is just like the difference between, on the one hand, a watch or other automaton (that is, a self-moving machine), when it is wound up and contains in itself the corporeal principle of the movements for which it was designed, together with everything else required for its operation; and on the other hand, the same watch or machine when it is broken and the principle of its movement ceases to be active.
>
> —(1649/1985, pp. 329–330)

Thus Descartes did not treat the immaterial soul as the source of the vitality of the material body or the departure of the immaterial soul as the cause of bodily death. According to Descartes, this common error likely arose from "supposing that since dead bodies are devoid of heat and movement, it is the absence of the soul which causes this cessation of movement and heat":

> Thus it has been believed, without justification, that our natural heat and all these movements of our bodies depend upon our soul; whereas we ought to hold, on the contrary, that the soul takes its leave when we die only because this heat ceases and the organs which bring about bodily movement decay.
>
> —(1649/1985, p. 329)

This account of biological functions in terms of an emergent **vital force** of organized matter stimulated a fertile tradition of physiological research, although it later became the object of criticism by reductive physiologists, notably in the 19th century.

Introspection and Images Descartes claimed that we have infallible **introspective knowledge**: that our conscious apprehension of our own mental states such as sensations, beliefs, emotions, thoughts, and memories is direct and certain. In developing his system of knowledge from first principles, Descartes argued that although he could doubt that sense experience provides knowledge of the existence and properties of material bodies in the external world, he could not doubt the contents of his sense experience—of how things appeared to his senses. Consequently, even if his judgment that he was sitting by a bright and crackling fire was false because he was dreaming this while asleep in bed, he could at least be certain that this was how things *appeared* to his consciousness (1641/1985, p. 19). In this view, as long as we restrict our judgments to the contents of our consciousness they are immune from error. We err only when we make inferences about material bodies in the external world on the basis of our sense experience.

The view that our knowledge of mental states is direct and certain was maintained by most psychologists and philosophers throughout the succeeding centuries and remained popular until the early decades of the 20th century. Empiricists such as John Locke (1632–1704), George Berkeley (1685–1753), and David Hume (1711–1776), who rejected Descartes' rationalist claims about innate ideas and the rational intuition of fundamental scientific principles and maintained that all our ideas and knowledge are derived from experience, also embraced this account of self-knowledge of mental states.

Descartes also articulated the problem about our knowledge of the external world that vexed later empiricists: What justification do we have for making inferences about the existence and properties of material bodies in the external world on the basis of our sense experience? How do we know that there are material bodies in the external world that have the colors and shapes that we attribute to them on the basis of sense experience? This was not a problem generated by the mere possibility of doubting the existence and properties of material bodies in the external world, but was a product of Descartes' treatment of thoughts and ideas as images.

Descartes characterized the problem about our knowledge of the existence and properties of material bodies in the external world as a problem about the justification of our belief that our ideas of material bodies and their properties *resemble* material bodies and their properties:

> But the chief question at this point concerns the ideas which I take to be derived from things existing outside me: what is my reason for thinking that these resemble these things?
>
> —(1641/1985, p. 26)

This was a serious problem for Descartes, since his commitment to the distinction between primary and secondary qualities forced him to acknowledge that it was doubtful if our ideas of secondary qualities resemble the real qualities of material bodies:

> There may be a difference between the sensation we have of light (i.e. the idea of light which is formed in our imagination by the mediation of our eyes) and what is in the objects that produces that sensation within us (i.e. what is in the flame or the sun that we call by the name "light"). For although everyone is commonly convinced that the ideas we have in our minds are wholly similar to the objects from which they proceed, nevertheless I cannot see any reason which assures us that this is so.
>
> —(1664/1985, p. 81)

Later empiricists such as Locke, Berkeley, and Hume shared Descartes' conception of thoughts and ideas as images. They also recognized that this caused a problem for the justification of our claims to have knowledge of the existence and properties of material bodies in the external world, since we cannot directly compare imagistic thoughts with external reality, in the way that we can compare a representational painting to the actual physical scene it is intended to represent (a painting of the Grand Canal in Venice with the actual Grand Canal in Venice, for example). This conception of thoughts as images remained popular for many centuries, and impeded the development of a psychology of thought until the early 20th century.

Nevertheless, Descartes also deserves credit for being one of the earliest theorists to recognize that thoughts cannot be equated with images. He noted that although we can conceive of both a triangle and a chiliagon (a figure with a thousand sides) and can form an image of a triangle, we cannot form an image of a chiliagon (1641/1985, p. 50).

La Mettrie: Machine Man

Descartes had taken the revolutionary step of extending the principles of mechanistic explanation to all animal and some human behavior by treating such behavior as the product of matter in motion, but had resisted extending these principles to the human mind, whose material basis he denied. The French military physician Julien Offroy de La Mettrie had no such qualms and boldly declared that "man is a machine" and "there is in the whole universe only one diversely modified substance" (1748/1996, p. 39).

A native of Brittany, La Mettrie received his medical education at the University of Leiden in Holland. He practiced as a physician in Leiden for a number of years, publishing papers on smallpox, venereal disease, and vertigo until commissioned as an army physician during the Franco-Austrian war. He is reputed to have developed his materialist views as a consequence of a fever contracted during the siege of Freiburg: The disorders of thought and emotion induced by the fever left a lasting impression on him. La Mettrie published *The Natural History of the Soul* in 1745, in which he argued that humans are complex animals. This work created such an uproar amongst the French clergy that La Mettrie was forced to return to Holland. In 1748 he produced his major work *Man Machine*, in which he argued that the principles of mechanistic explanation should be extended to all human behavior, including human thought and language. When the blatant materialism and implicit atheism of this work proved too much even for the enlightened Dutch, La Mettrie moved to Berlin at the invitation of Frederick the Great, who became his biographer. There he died prematurely, through hedonistic overindulgence, according to his meaner critics. He expired during a bout of indigestion brought on by a surfeit of pheasant and truffles.

Organized Matter La Mettrie believed that the organization of matter held the key to the understanding of all animal and human behavior:

> Since all the soul's faculties depend so much on the specific organization of the brain and of the whole body that they are clearly nothing but that very organization, the machine is perfectly explained!
>
> —(1748/1996, p. 26)

He claimed that "organized matter is endowed with a motive principle, which alone distinguishes it from unorganized matter" and that the gradations of complexity of animal and human behavior are "dictated by the diversity of this organization" (1748, p. 33). He consequently maintained that human thought is an emergent property of matter at a complex level of organization:

> I believe thought to be so little incompatible with organized matter, that it seems to be one of its properties, like electricity, motive power, impenetrability, extension, etc.
>
> —(1748/1996, p. 35)

The extension of mechanistic explanation to thought was so obvious and natural, according to La Mettrie, that Descartes must have been convinced of it. Although Descartes, who "understood animal nature and was the first to demonstrate perfectly that animals were mere machines," publicly avowed that mind and body are distinct substances, "it is obvious that it was only a trick, a cunning device to make the theologians swallow the poison hidden behind an analogy that strikes everyone and that they alone cannot see":

> For it is precisely that strong analogy which forces all scholars and true judges to admit that, however much these haughty, vain beings . . . may wish to exalt themselves, they are basically only animals and vertically crawling machines.
>
> —(1748/1996, p. 35)

La Mettrie claimed that the man machine is materially continuous with the animal machine: "From animals to man there is no abrupt transition" (1748/1996, p. 13). He maintained that the man machine differs from the animal machine only in terms of the degree of complexity of its material organization:

> We can see that there is only one substance in the universe and that man is the most perfect one. He is to the ape and the cleverest animals what Huygen's planetary clock is to one of Julien Leroy's watches.
>
> —(1748/1996, pp. 33–34)

La Mettrie held that physicians were "the only natural philosophers who have the right to speak on this subject" (1748, p. 5), since only their views were based upon "experience and observation alone" (1748, p. 4). He offered two forms of evidence in support of his materialist theory of mind and his claims about the continuity between animal and human machines. He documented the effects of various ingested substances, such as opium, wine, coffee, and red meat upon human thought and emotion and noted how damage caused to the "springs" of the human machine by fever or poisoning can produce severe disruption to mental functioning in the form of delusions and mania. He also appealed to the studies in "comparative anatomy" conducted by the Oxford neuroanatomist Thomas Willis (1621–1675), author of *The Anatomy of the Brain* (1664) and *Two Discourses Concerning the Soul of Beasts* (1672):

> In general, the form and composition of the quadruped's brain is more or less the same as man's. Everywhere we find the same shape and the same arrangement, with one essential difference: man, of all the animals, is the one with the largest and most convoluted brain, in relation to the volume of his body. Next come the ape, beaver, elephant, dog, fox, cat, etc.: these are the animals that are most like man, for we can also see in them the same graduated analogy concerning the corpus callosum.
>
> —(1748/1996, pp. 9–10)

However, La Mettrie's appeal to the effects of ingested substances, fever, and poisoning on mental functioning hardly established materialism. Although he demonstrated that many "states of the soul are . . . related to those of the body" (1748/1996, p. 9), regular correlation between mental and bodily states was entirely consistent with Descartes' interactionist dualism and was in fact presupposed by it. Although the evidence from comparative anatomy supported the hierarchical gradation of human and animal psychology and behavior, it did not demonstrate their continuity. Many ancient and medieval theorists acknowledged the hierarchical gradation of humans and animals, but maintained that some psychological capacities, such as abstract thought and language, are attributable only to humans.

Strong and Weak Continuity La Mettrie avowed two forms of continuity between humans and animals, which should be carefully distinguished. One is the **weak continuity** between humans and animals presupposed by materialism: the notion that humans and animals, like vegetables and minerals, are composed of the same basic material, differently organized. La Mettrie rejected Descartes' claim that humans and animals are fundamentally discontinuous because humans have an immaterial mind and animals do not. As he put it,

> Man is not molded from a more precious clay; nature has used one and the same dough, merely changing the yeast.
>
> —(1748/1996, p. 20)

Thus he maintained that human thought is materially instantiated in the human brain.

However, La Mettrie also argued for **strong continuity** between human and animal psychology and behavior. He claimed that differences between human and animal psychology and behavior are merely differences in degree and not fundamental differences in kind. According to this view, human psychology and behavior are fundamentally *identical* to animal psychology and behavior. Human psychology and behavior may be re-identified in other animals, albeit in attenuated form, since human psychology and behavior are merely more complex forms of animal psychology and behavior. Thus La Mettrie argued that thought and language could be attributed to animals, although in attenuated form.

However, the weak continuity of materialism does not entail the strong continuity of human and animal psychology and behavior, any more than the weak continuity of the inorganic and organic presupposed by materialism entails strong continuity of structure and function between the inorganic and organic. Rocks and plants are both composed of organized matter, but plants have properties, such as the ability to perform photosynthesis, that are not instantiated *to any degree* in rocks. Consequently, although humans and animals are both composed of organized matter, it might still be the case that humans have some psychological capacities, such as the capacity for abstract thought or language, that are not instantiated *to any degree* in animals. Whether or not the capacity for abstract thought or language is in fact instantiated in animals is a separate empirical matter.

The weak continuity of materialism and strong continuity between human and animal psychology and behavior became associated historically because Descartes rejected both materialism and strong continuity and because most later evolutionary theorists, comparative psychologists, and behaviorists were materialists who, like La Mettrie, affirmed strong continuity. However, there is no intrinsic connection between materialism and strong continuity. Aristotle and other ancient and medieval theorists affirmed the weak continuity of materialism but denied strong continuity between human and animal psychology and behavior, as did the comparative psychologist Conwy Lloyd Morgan and the behaviorist psychologist John B. Watson, who were both materialists but maintained that only humans have the capacity for language.

Animals and Language Although it was not mandated by his materialism, La Mettrie maintained that human and animal psychology and behavior are strongly continuous. He acknowledged that only humans speak a language, but denied that machines and animals are incapable of learning a language. He believed that language is the product of intelligence and learning, which he held to be a function of brain size. Given the anatomical and behavioral similarities between apes and humans, La Mettrie was convinced that apes are capable of learning language:

> The similarity of the ape's structure and functions is such that I hardly doubt at all that if this animal were perfectly trained, we would succeed in teaching him to utter sounds and consequently to learn a language. Then he would no longer be a wild man, nor an imperfect man, but a perfect man, a little man of the town, with as much substance or muscle for thinking and taking advantage of his education as we have.
>
> —(1748/1996, p. 12)

He suggested teaching apes language using the techniques developed by J. C. Amman (1700/1965) for teaching sign language to deaf-mutes. In the 20th century, Allen and Beatrice Gardner used similar techniques to teach the sign language of the deaf to chimpanzees (Gardner & Gardner, 1969).

La Mettrie claimed that the linguistic competencies of humans, like their developed forms of social and cultural behavior, are based upon interpersonal imitation or "mimicry," a form of reflexive learning that is as automatic as the pupillary reflex:

> We take everything—gestures, accents, etc.—from those we live with, in the same way as the eyelid blinks under the threat of a blow that is foreseen, or as the body of a spectator imitates mechanically, and despite himself, all the movements of a good mime.
>
> —(1748/1996, p. 9)

He maintained that animals are also capable of imitation or "mimicry," and noted how a monkey can learn "to put on and take off his little hat or to ride a trained dog" (1748/1996, p. 13). Similar accounts of imitative learning formed the basis of the theories of social behavior developed by Gustav Le Bon (1841–1931) and Gabriel Tarde (1843–1904) in the late 19th century, which played a major role in shaping the development of 20th-century American social psychology. Like 20th-century behaviorist psychologists, La Mettrie believed that the same basic principles of learning applied to animals and humans and that these principles could be exploited to improve their condition through training and education.

God, Nature, and Morality Although La Mettrie affirmed the probability of the existence of a "supreme Being" (1748/1996, p. 22), he denied that it vouchsafed the doctrines of any established religion. He was scornful of academic arguments for the existence of God, particularly those based upon the diversity and functional adaptation of animal species, and the apparently purposive nature of biological development. He acknowledged that it was unlikely that such features were the product of blind chance, but claimed that "destroying chance does not prove the existence of a supreme Being" (1748/1996, p. 24). La Mettrie suggested another alternative, that functionality and apparent teleology are simply a product of the ordered development of nature itself:

> The eye sees only because it happens to be organized and placed as it is; and that, given the same rules of movement followed by nature in the generation and development of bodies, it was not possible for that wonderful organ to be organized and placed otherwise.
>
> —(1748/1996, p. 25)

Certainly La Mettrie took seriously the possibility that there is no purpose or design informing human existence:

> Who knows after all whether the reason for man's existence is not his existence itself. Perhaps he was thrown by chance on a point on the earth's surface without being able to say how or why, but simply that he has to live and die, like mushrooms which appear from one day to the next, or flowers which grow beside ditches and cover walls.
>
> —(1748/1996, p. 23)

Such an uncompromising materialist and mechanistic conception appeared to paint a very bleak picture of human nature. It suggested that humans are no better than animals, concerned only with the satisfaction of sensual desires, especially given La Mettrie's celebration of the sexual nature of the human machine. This was precisely the consequence of treating men as machines that Descartes had feared.

Yet La Mettrie was rather more sanguine about the prospects for humanity. He questioned the common assumption that humans are morally superior to animals, noting that animals rarely murder or torture each other or engage in religious wars and claimed that some animals are capable of moral emotions such as remorse. More significantly, he stressed that a materialist and mechanistic account of human thought and behavior does not preclude human virtue, since it treats human virtue as a product of material organization on a par with thought and digestion:

> Since thought clearly develops with the organs, why should not the matter that composes them not also be capable of remorse once it has acquired, with time, the faculty of feeling? . . .
>
> Given the slightest principle of movement, animate bodies will have everything they need to move, feel, think, repent, and in a word, behave in the physical sphere and in the moral sphere which depends upon it.
>
> —(1748/1996, p. 26)

According to La Mettrie, there is no special reason to suppose that human machines would pursue their own selfish interests at the expense of others. On the contrary:

> The materialist, convinced, whatever his vanity might object, that he is only a machine or an animal, will not ill-treat his fellows. . . . Following the law of nature given to all animals, he does not want to do to others what he would not like others to do to him.
>
> —(1748/1996, p. 39)

Although La Mettrie's work had a major impact in the 18th century, his name became associated with such odium that he was rarely cited and consequently had little effect on the later development of psychology. Although his commitment to strong continuity presaged a fundamental principle of evolutionary theory and behaviorist psychology, later evolutionists and behaviorist psychologists seem to have been unaware of his work. When Thomas Huxley (1825–1895) addressed the British Association in Belfast in 1874 on "The Hypothesis That Animals Are Automata, and Its History," he gave due credit to Descartes but made no mention of La Mettrie (Boakes, 1984).

Thomas Hobbes: Empiricism, Materialism, and Individualism

The Englishman Thomas Hobbes (1588–1679) shared La Mettrie's materialist vision of human psychology but took a rather more pessimistic view of its implications. Born in Malmesbury, England, Hobbes was educated at Oxford University and served as Francis Bacon's secretary for a short period. He entered the employment of William Cavendish, third Earl of Devonshire, and for most of the rest of his life served as secretary and tutor to the family. This put him in some danger during the period leading up to the English Civil War. Hobbes fled to France in 1640 and did not return until 1651. He made several tours of Europe, where he met many of the leading theorists of his day, such as Galileo and Descartes, with whom he became friends. He lived to age 91, producing translations of Homer's *Iliad* and *Odyssey* at age 87.

Hobbes's main interests were political, and his major work *Leviathan* (1651) is primarily an argument in favor of absolute monarchy. His aim was to devise a political system capable of avoiding the horrors of civil war, having been greatly affected by the English Civil War, albeit at a distance. His psychological theories are mainly to be found in *On Human Nature* (1640) and the preliminary chapters of *Leviathan*.

Hobbes claimed that his psychological interests were aroused after having read Euclid's *Elements* at age 40; it induced his reverence for the self-contained axiomatic systems of geometry. Consequently, he tried to deduce his claims about human psychology and society from a number of self-evident axioms, based upon the principles of the new mechanistic science. Although Hobbes followed Descartes in adopting a deductive approach to explanation, he rejected Descartes' rationalist account of knowledge and denied the existence of innate ideas.

Hobbes was a psychological empiricist who maintained that all our ideas or concepts are derived from sense experience:

> The original of them all is that which we call *sense*; for there is no conception in a man's mind which hath not at first, totally, or by parts, been begotten upon the organs of sense. The rest are derived from that original.
>
> —(1651/1966, p. 1)

Hobbes was also committed to the **homogeneity of cognition and sense perception**: He claimed that the difference between cognition and sense perception is a matter of degree (of intensity), but not a fundamental difference in kind. On this account, thinking of a tree in blossom is like seeing and smelling a tree in blossom, only fainter. Hobbes agreed with Descartes that ideas are like pictorial images, since he maintained, with later empiricists, that our ideas are copies or faint images of sense impressions of objects:

> For after the object is removed, or the eye shut, we still retain an image of the thing seen, though more obscure than when we see it. . . . *Imagination* therefore is nothing more than *decaying sense*.
>
> —(1651/1966, p. 4)

Hobbes derived his first principles from the materialism of the new mechanistic science. Adopting a reductive explanatory approach to human psychology, he claimed that mental states and processes are "nothing really, but motion in some internal substance of the head":

> which motion not stopping there, but proceeding to the heart, of necessity must either help or hinder the motion which is called vital; when it helpeth, it is called delight, contentment, or pleasure, which is nothing really but motion about the heart, as conception is nothing but motion in the head; and the objects that cause it are called pleasant or delightful.
>
> —(1640/1966, p. 31)

Hobbes also embraced a form of **psychological hedonism**, according to which all human behavior is determined by the desire to attain pleasure and avoid pain:

> This motion, in which consisteth pleasure or pain, is also a solicitation or provocation either to draw near the thing that pleaseth, or to retire from the thing that displeaseth; and this solicitation is the endeavor or internal beginning of animal motion, which when the object delighteth, is called appetite; and when it displeaseth, it is called aversion.
>
> —(1640/1966, p. 31)

Hobbes famously claimed that the unbridled pursuit of selfish interest would inevitably lead to the war "of every man, against every man" and that in such a "state of nature" the life of man would be "solitary, poor, nasty, brutish, and short" (1651/1966, p. 113). He believed that humans embrace systems of civic government out of self-interest, in order to avoid these anticipated consequences. He argued that an absolute monarchy, in which individuals abandon their rights to a sovereign power, is the most just and efficient form of government, although he maintained that any form of government is better than none.

Frontispiece of Hobbes's *Leviathan* (1651). Body of state represented as aggregation of individual persons.

Hobbes's explanatory reductionism is also manifest in his **individualism**, which formed the basis of his account of social community in *Leviathan*. According to Hobbes, societies or social groups are nothing more than collections of human individuals, and social behavior is nothing more than the aggregate behavior of collections of human individuals, determined by their pursuit of pleasure and avoidance of pain. This individualistic conception of the social was characteristic of later empiricist concepts of the social, from Adam Smith (1723–1790) to Floyd Allport (1890–1978), who determined the course of American social psychology in the early 20th century (Katz, 1991).

In advancing these materialist and mechanistic explanations of human psychology and behavior, Hobbes denied that humans have free will. He claimed that the "will" is just the most powerful appetite, or efficient cause, and is the same in animals and men. He thus reduced final causation in the realm of human psychology and behavior to efficient causation:

> A final cause has no place but in such things as have sense and will, and this also I . . . prove . . . to be an efficient cause.
>
> —(1655/1966, p. 132)

Like La Mettrie, Hobbes was condemned by the religious establishment for his materialist views. He was denied admission to the newly formed Royal Society, which is perhaps not that surprising, since although he was a vigorous advocate of the new mechanistic science, he did not make any substantive contribution to it. He did, however, take the first step in extending mechanistic forms of explanation to mental processes. He offered tentative explanations of "trains of thought," likening the "coherence" of thought to the "coherence" of matter. He suggested that ideas derived from sense experience are connected in our memory by their conjunction in our sense experience:

> The *cause* of the *coherence* or consequence of one conception to another, is their first *coherence* or consequence at that time when they are produced by *sense*.
>
> —(1640/1966, p. 15)

Hobbes is sometimes treated as the father of British empiricism and the founder of what later came to be known as associationist psychology. Yet although he was the first to clearly articulate many of the distinctive principles of British empiricism, such as the principles of psychological empiricism and the homogeneity of cognition and sense perception, and did suggest a mechanistic treatment of the association of thought, his contribution was more programmatic than substantive. It was John Locke who detailed the origin of complex ideas in sense experience and David Hume and David Hartley (1705–1757) who developed the principles of association that grounded the later development of associationist psychology.

MENTAL MECHANISM AND STIMULUS-RESPONSE PSYCHOLOGY

By the end of the 17th century the triumph of mechanism was complete in the physical sciences, and mechanistic explanation was extended to human psychology and behavior in the 18th and 19th centuries. However, this did not lead to a progressive acceptance of materialism, as might have been expected. Most of those who developed mechanistic explanations of mental states and processes did their best to avoid any association with materialism. Even those who explored the material basis of mentality in the brain avowed a form of dualism or a **neutral parallelism**, by maintaining that every mental state is correlated with a brain state, while carefully avoiding speculation about the basis of the correlation between mental and brain states. Although the power of organized religion declined over these centuries, the religious establishment still played a powerful role within society and civic administration, often determining royal or government patronage and university positions.

One of the peculiarities of Descartes' pioneering account of reflexive behavior in animals and humans was that he presumed that the nerves from sensory receptors are connected in the brain to the nerves that control motor behavior, even though it was common knowledge that animals often continue to display reflexive behavior after decapitation. For example, La Mettrie noted how

> A drunken soldier cut off the head of a turkey-cock with a sabre. The animal stayed upright, then it walked and ran; when it hit the wall it turned around, beat its wings, still running, and finally fell down.
>
> —(1748/1996, p. 27)

The English clergyman Stephen Hales (1677–1761) demonstrated that decapitated frogs continue to respond to stimulation so long as the marrow of their spinal cord remains intact (La Mettrie's *Man Machine* was provocatively dedicated to Hales). Robert Whytt (1714–1766), who taught in the medical school at the University of Edinburgh, confirmed these results in a careful series of experiments:

> When any of the muscles of the leg of a frog are irritated some time after cutting off its head, almost all the muscles belonging to the legs and thighs are brought into contraction, if the spinal marrow be entire.
>
> —(cited in Smith, 1992, p. 74)

He also noted that decapitation enhances reflexive activity (Smith, 1992) and that some reflexes can be preserved even if only a small portion of the spinal cord remains intact (Boakes, 1984).

Whytt maintained that such experiments demonstrated that

> a certain power of influence lodged in the brain, spinal marrow, and nerves, is either the immediate cause of the contraction of muscles of animals, or at least necessary to it.
>
> —(1751/1978, p. 3)

He claimed that decapitated animals respond selectively to stimulation and noted how a brainless frog will use its legs to relieve the irritation caused by an acid-soaked tissue applied to its skin, just as many intact animals use their legs to rid themselves of fleas and ticks (Reed, 1997).

Whytt suggested that such experiments demonstrated the existence of an unconscious "sensitive soul" in the spinal cord, capable of making adaptive responses to sensory stimulation. Like Thomas Willis, the Oxford neuroanatomist, he suggested that mentality is distributed throughout the nervous system and not restricted to the brain. This suggestion generated opposition as fierce as that for La Mettrie's claim that mentality is a property of the brain. However, it indirectly stimulated many 19th-century neurophysiologists, who often saw themselves as opponents of such crass materialism, to locate mentality in the higher regions of the brain, such as the cerebral cortex.

In his *Essay on the Vital and Other Involuntary Motions of Animals* (1751), Whytt identified a range of innate reflexive behavior, such as digestion, coughing, sneezing, and penile erection (Boakes, 1984). He introduced the notion of a **stimulus** into the theoretical vocabulary, defined as the application of any form of physical energy to a nerve (Reed, 1997). He also noted how certain originally neutral stimuli can acquire the capacity to generate innate reflexes by association with their precipitating stimuli (Boakes, 1984), anticipating Pavlov's account of conditioned reflexes, including the form of conditioned salivation that became the primary focus of his experimental studies:

> Thus the sight, or even the recalled idea of grateful food causes an uncommon flow of spittle into the mouth of a hungry person; and the seeing of a lemon cut produces the same effect in many people.
>
> —(cited in Boakes, 1984, p. 95)

DISCUSSION QUESTIONS

1. Bacon claimed that scientific theories should be judged by the utility of their "works" or "discoveries" and that genuine scientific knowledge leads to "dominion" over nature. What useful works or discoveries have scientific psychological theories promoted? Is dominion over humans an appropriate goal for scientific psychology? If so, to what degree? In what sense?
2. Descartes believed that animals lack sensory awareness and consciousness. Do you? How could you tell? Can you think of a "crucial instance" or "crucial experiment" that would demonstrate sensory awareness or consciousness in animals? Could a machine have sensory awareness or consciousness? How could you tell?
3. Does thinking of animals as machines incline us to think they are more or less likely to be capable of language and problem solving?
4. Does materialism imply strong continuity between human and animal psychology and behavior?
5. Hobbes was an individualist about social community. Are social attitudes and behavior just the common attitudes and behavior of a collection of individuals? Is contemporary social psychology individualist, or does it conceive of social attitudes and behavior as something more than (or different from) the aggregation of attitudes and behavior?

GLOSSARY

automaton A moving machine.

corpuscle Seventeenth-century term for the atomistic components of material bodies, coined by Robert Boyle.

corpuscularian theory of light Theory of light in which it is treated as a stream of material corpuscles, or atoms.

epicycles A system of circles within circles introduced by Ptolemy (and Copernicus) to accommodate the "wandering" motion of planets.

geocentric theory The theory that Earth is the fixed center of the universe, around which the sun and other planets orbit.

heliocentric theory The theory that the sun is the fixed center of the universe, around which Earth and other planets orbit.

homogeneity of cognition and sense perception The claim that cognition and sense perception differ in degree (of intensity) but not fundamental kind, usually via the claim that ideas are weaker images of sense impressions.

Idols of the cave Cognitive biases in scientific thinking that are idiosyncratic products of individual human development.

Idols of the marketplace Social biases in scientific thinking based upon notions derived from common linguistic usage.

Idols of the theatre Social biases in scientific thinking based upon theories maintained by schools of philosophy as received dogma.

Idols of the tribe Cognitive biases in scientific thinking based upon innate human propensities.

individualism The view that societies or social groups are nothing more than collections of human individuals and that social behavior is nothing more than the aggregate behavior of collections of human individuals.

Instance of the Fingerpost Francis Bacon's name for a crucial instance that enables the empirical adjudication of competing theories.

interactionism The view that mind and body causally interact.

introspective knowledge The conscious apprehension of mental states, usually held to be direct and certain.

mechanistic explanation Efficient causal explanation in terms of antecedent conditions sufficient to produce an effect, often associated with a conception of the universe as a giant (usually clockwork) machine.

neutral parallelism The view that every mental state is correlated with a brain state, without commitment to any theory about the nature of the relation between mental and brain states.

occasionalism The view that God directly causes the regular correlation between mental and bodily states.

pre-established harmony The view that God maintains the regular correlation between mental and bodily states.

psychological hedonism The view that all human behavior is motivated by the desire to attain pleasure and avoid pain.

reflexive behavior Automatic and involuntary behavior in response to stimulation.

Reformation The Protestant religious movement founded by Martin Luther.

Renaissance The cultural movement that began in southern Italy in the 14th century and promoted innovative developments in art, literature, architecture, and music, as well as in mathematics, religion, and science.

Renaissance humanism The Renaissance focus on human psychology and celebration of its potential.

stellar parallax The variation in the angular separation of the stars that was a crucial implication of the Copernican heliocentric theory.

stimulus Term introduced by the Edinburgh physician Thomas Whytt to describe the application of any form of physical energy to a nerve.

strong continuity The view that the differences between human and animal psychology and behavior are differences in degree and not fundamental differences in kind.

vital force An emergent force of organized matter held to explain biological functions such as bodily heat and movement.

vortex theory of motion Descartes' theory of motion in terms of "action by contact."

weak continuity The view that humans and animals are composed of the same basic material, differently organized.

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CHAPTER 5

The Newtonian Psychologists

THE ACHIEVEMENTS OF THE SCIENTIFIC REVOLUTION REPRESENTED THE vanguard of the **Enlightenment**, that period in European thought in the 17th and 18th centuries when confidence in human reason and experience gradually came to supersede faith in religion and traditional authority. One central feature of Enlightenment thought, which flourished in France, Scotland, England, and Germany, was a commitment to human progress and an optimistic belief in the applicability of scientific knowledge, including social and psychological knowledge, to the improvement of the human condition. The Enlightenment saw the emergence of more liberal, secular, and utilitarian concepts of humanity and the development of more democratic societies, such as the United States. Although not universally embraced, these Enlightenment ideals continue to inform contemporary confidence in the theoretical progress and social utility of the sciences, including social and psychological science.

The rejection of the Aristotelian tradition was good news for the natural sciences. The rejection of Aristotle's geocentric theory and final causal explanations of motion led to advances in astronomy and physics. However, it was not so obviously good news for psychology. One of the casualties of the scientific revolution was Aristotle's biologically grounded functional psychology, which came to be replaced by a variety of mechanistic psychological theories. This was not the intent of the pioneers of the new science, such as Galileo, Bacon, and Newton. Although they maintained that final causal explanation has no place in physical science, most recognized that final causal explanation is entirely appropriate in the realm of human and animal behavior. Yet this qualification was generally ignored by the protopsychologists of the 17th and 18th centuries, who tried to create a science of psychology based upon the mechanistic forms of efficient causal explanation characteristic of the new science, for which Newton's physics came to serve as a paradigm.

Newton's theory of universal gravitation was hugely influential, not only in physical science, where it continued to reign supreme throughout the 18th and 19th centuries, but also with respect to the forms of psychological theory that developed during these centuries. These were either attempts to model psychological theory upon Newtonian science, such as associationist psychology, or